QPCODE **Reg. No**

: .....

Name

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MAHATMA GANDHI UNIVERSITY , KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

SEMESTER III

Minor- DSC B: MG3DSCPHY202: Atomic and Molecular Spectroscopy

(2024 ADMISSION ONWARDS)

Duration: 1.5 hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)		
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.		
Part A		
Short Answer Type Questions		
Answer any **10** questions		
Each question carries **2** marks		
1.	If the Rydberg constant is 1.097 x10^{7} m^{-1}, find the wave number of the Balmer series limit.	[U] / [1]
2.	Define gyromagnetic ratio	[K] / [1]
3.	How does spin angular momentum arise?	[U] / [1]
4.	What is Paschen Back effect?	[K] / [1]
5.	What does the term symbol $^{2}P_{3/2}$ represent?	[U] / [1]

6.	What is Zeeman effect?	[K] / [1]
7.	State mutual exclusion principle.	[K] / [2]
8.	Draw the first 4 energy levels of a rigid rotator	[U] / [2]
9.	What do you understand by a spherical top molecule?	[U] / [2]
10.	What is zero point energy?	[U] / [2]
11.	Among Electronic and Vibrational spectra which spectra comes in the visible range? Explain	[U] / [2]
12.	What is ESR?	[K] / [3]
13.	What is the principle of NMR?	[K] / [3]
14.	Why does ESR be used to study paramagnetic substances?	[U] / [3]
		[2x10 = 20]
	Part B	
	Short Essay Type Questions	
	Answer any **6** questions	
	Each question carries **5** marks	
15.	State and Explain the postulates of Bohr atom model.	[U] / [1]
16.	Briefly give an idea about the quantum numbers in vector atom model.	[U] / [1]
17.	Differentiate between LS and JJ coupling	[U] / [1]

18.	Explain the origin of the D-lines of sodium.	[U] / [1]
19.	Compare Stokes and anti-Stokes lines.	[U] / [2]
20.	Write the differences between fluorescence and phosphorescence.	[U] / [2]
21.	Give a brief idea about various types of molecular energies.	[K] / [2]
22.	Explain the resonance condition for NMR	[U] / [3]
		[5x6 = 30]